

# ML-Enabled E-Governance Citizen Chatbot

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## 1. Project Objective

The project aims to develop a citizen-friendly chatbot that assists users in understanding and accessing Tamil Nadu e-Governance/e-Sevai services. The system uses machine learning-based service classification and Retrieval-Augmented Generation (RAG) to retrieve relevant government service information and provide concise answers to citizens.

## 2. Data Extraction Prompts

### 2.1 Government Service Data Extraction Prompt

You are a data extraction assistant for an e-Governance Citizen Chatbot.

Extract the following information from the official government webpage/manual provided below:

1. Service code
2. Service name
3. Department
4. Simple description of the service
5. Eligibility
6. Documents required
7. Where to apply
8. How to apply
9. Service charge
10. Processing time

## 11. Official source

Rules:

- Use only information present in the provided official government source.
- Do not guess, assume, or invent information.
- If a field is not available, write "Not specified in the available official source".
- Preserve fees and charges as stated in the source.
- Keep the description simple and suitable for a citizen chatbot.
- List each required document separately.
- Return the extracted information clearly.

Official government source:

<https://www.tnesevai.tn.gov.in/citizen/Pages/VideoTutorial.aspx>

<https://tnesevai.tn.gov.in/Pages/EsevaiServiceList.aspx>

## 2.2 JSON Conversion Prompt

Convert the extracted government service information into valid JSON using exactly this structure:

```
{  
  "service_code": "",  
  "service_name": "",  
  "department": "",  
  "description": "",  
  "documents_required": [],  
  "where_to_apply": [],  
  "how_to_apply": "",  
  "service_charge": "",  
  "processing_time": "",  
  "official_source": ""  
}
```

Rules:

- Keep the field order exactly as shown.

- Do not add extra fields.
- Do not remove fields.
- Do not invent missing information.
- If information is unavailable, use "Not specified".
- Keep documents as an array.
- Keep where\_to\_apply as an array.
- Keep the official government URL in official\_source.
- Return JSON only.

### **3. Common e-Sevai Information Extraction Prompt**

Extract the common e-Sevai information required for a citizen guidance chatbot.

Extract information about:

#### **1. CAN**

- \* What is CAN?
- \* How to create CAN?
- \* CAN validity
- \* Forgotten CAN

#### **2. Login and Registration**

- \* Registration procedure
- \* Login procedure
- \* Required details

#### **3. Application Status**

- \* How to check application status
- \* Information required to check status

#### **4. Grievances and Complaints**

\* How to submit a grievance or complaint

## 5. Helpdesk

\* Email

\* Phone number

\* Other official contact information

### Rules:

- Use only the official Tamil Nadu e-Sevai source provided.
- Do not guess or add information.
- If information is unavailable, write "Not specified".
- Keep the information short and suitable for a citizen chatbot.
- Include the official source for each section.
- Return valid JSON only.

## 4. Official Government Sources

The following official Tamil Nadu e-Sevai sources were used for collecting the information:

Tamil Nadu e-Sevai Citizen Portal – Video Tutorials

<https://www.tnesevai.tn.gov.in/citizen/Pages/VideoTutorial.aspx>

Tamil Nadu e-Sevai Service List

<https://tnesevai.tn.gov.in/Pages/EsevaiServiceList.aspx>

Tamil Nadu e-Sevai FAQ

<https://www.tnesevai.tn.gov.in/Pages/FAQ.aspx>

## **5. Extracted Common e-Sevai Information**

### **5.1 Citizen Access Number (CAN)**

#### **What is CAN?**

CAN stands for Citizen Access Number. It is required to avail e-Sevai services offered through the Tamil Nadu e-Governance system.

#### **CAN Validity**

CAN has lifetime validity.

#### **Forgotten CAN**

A forgotten CAN can be searched using:

- Mobile Number
- Date of Birth
- Name
- Email ID

#### **Official Source**

<https://www.tnesevai.tn.gov.in/Pages/FAQ.aspx>

### **5.2 Login and Registration**

New users can register through the e-Sevai Citizen Portal.

The citizen portal provides registration and login facilities for accessing e-Sevai services.

#### **Login Details**

The login page requires:

- User Name
- Password
- Captcha

#### **Official Source**

<https://www.tnesevai.tn.gov.in/citizen/>

### **5.3 Application Status**

Citizens can check the status of an application after logging into the e-Sevai Citizen Portal.

**Procedure**

- Log in to the e-Sevai Citizen Portal.
- Select Check Status.
- Enter the Application Number.
- Check the application status.

**Information Required**

- Application Number

**Official Source**

<https://www.tnesevai.tn.gov.in/Pages/FAQ.aspx>

**5.4 Grievances and Complaints**

The Tamil Nadu e-Sevai portal provides a Grievances/Complaints facility for citizens.

Detailed submission information should be taken only from the official source where available.

**Official Source**

<https://www.tnesevai.tn.gov.in/citizen/Pages/VideoTutorial.aspx>

**5.5 Helpdesk****Email**

[tnesevaihelpdesk@tn.gov.in](mailto:tnesevaihelpdesk@tn.gov.in)

**Toll-Free Phone Number**

18004256000

**Official Address**

TamilNadu e-Governance Agency  
807, 7th Floor,  
PT Lee Chengalvarayan Naicker Building,  
Anna Salai (Mount Road),  
Chennai – 600 002.

**Official Source**

<https://www.tnesevai.tn.gov.in/Pages/FAQ.aspx>

## 6. JSON Structure Used for Service Data

The extracted service information is converted into JSON using the following structure:

```
{
  "service_code": "",
  "service_name": "",
  "department": "",
  "description": "",
  "documents_required": [],
  "where_to_apply": [],
  "how_to_apply": "",
  "service_charge": "",
  "processing_time": "",
  "official_source": ""
}
```

The structure contains only the required fields and does not introduce additional fields.

## 7. Example of Extracted Service Data

### Community Certificate

```
{
  "service_code": "REV-101",
  "service_name": "Community Certificate",
  "department": "Revenue",
  "description": "Not specified",
  "documents_required": [],
  "where_to_apply": [],
  "how_to_apply": "Not specified",
  "service_charge": "₹60",
  "processing_time": "Not specified",
  "official_source": "https://tnesevai.tn.gov.in/Pages/EsevaiServiceList.aspx"
}
```

## Income Certificate

```
{
  "service_code": "REV-103",
  "service_name": "Income Certificate",
  "department": "Revenue",
  "description": "Not specified",
  "documents_required": [],
  "where_to_apply": [],
  "how_to_apply": "Not specified",
  "service_charge": "₹60",
  "processing_time": "Not specified",
  "official_source": "https://tnesevai.tn.gov.in/Pages/EsevaiServiceList.aspx"
}
```

Note: Fields such as eligibility, documents, processing time, and detailed procedure should be populated only when supported by the corresponding official government manual/source.

## 8. ML Classification and Hyperparameter Tuning

The chatbot uses TF-IDF for converting citizen questions into numerical text features and Logistic Regression for predicting the relevant e-Sevai service.

### Hyperparameters Tuned

Two Logistic Regression hyperparameters were varied:

#### 1. C

The following values were tested:

- 0.01
- 0.1
- 1
- 10
- 100

#### 2. Class Weight

The following options were tested:

- None
- Balanced

Therefore:

5 C values  $\times$  2 class-weight options = 10 configurations

## 9. Evaluation Metrics

Each hyperparameter configuration was evaluated using:

### Accuracy

Measures the proportion of correctly classified queries.

### Precision

Measures how many of the services predicted by the classifier were correct.

### Recall

Measures how many of the actual service classes were correctly identified.

### F1 Score

The F1 Score combines precision and recall and is used to select the best hyperparameter configuration.

For the project, weighted averages are used for Precision, Recall and F1 Score.

## 10. Hyperparameter Tuning Results

The following table represents the configurations evaluated by the Logistic Regression model.

|      |          |       |       |       |       |
|------|----------|-------|-------|-------|-------|
| 0.01 | None     | 0.252 | 0.289 | 0.252 | 0.226 |
| 0.01 | Balanced | 0.935 | 0.925 | 0.935 | 0.926 |
| 0.1  | None     | 0.822 | 0.762 | 0.822 | 0.78  |
| 0.1  | Balanced | 0.935 | 0.927 | 0.935 | 0.926 |
| 1    | None     | 0.935 | 0.918 | 0.935 | 0.921 |
| 1    | Balanced | 0.935 | 0.933 | 0.935 | 0.93  |
| 10   | None     | 0.944 | 0.942 | 0.944 | 0.938 |
| 10   | Balanced | 0.935 | 0.933 | 0.935 | 0.93  |

|      |          |       |       |       |       |
|------|----------|-------|-------|-------|-------|
| 100  | None     | 0.944 | 0.942 | 0.944 | 0.938 |
| 100  | Balanced | 0.935 | 0.933 | 0.935 | 0.93  |
| 0.01 | None     | 0.252 | 0.289 | 0.252 | 0.226 |